MILITARY SPECIFICATION

MODIFICATION KIT MK-710( )/TMQ (UNITS OF)

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fabricating units of Modification Kit MK-710( )/TMQ.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-654	- Brazing Alloy, Silver.
QQ-S-571	- Solder, Tin Alloy, Tin-Lead Alloy And Lead Alloy.

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-C-4150	- Case, Transit And Storage, Waterproof And Water-Vaporproof.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronic Equipment And Parts, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures For Packaging Materials.

MILITARY

MIL-STD-105	-	Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	-	Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-726	-	Packaging Requirements Code.

2.1.2 Other Government documents. The following Government drawings form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-DL-380825	-	Modification Kit MK-710( )/TMQ.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The modification kit MK-710( )/TMQ (hereinafter called "equipment") for Hydrogen Generator Set AN/TMQ-3 is designed to increase the volume of gas which can be produced in a single generation cycle, to cool the gas and reduce the moisture content of the generated gas. A coil condenser, with collection tank, is connected by hose between the hydrogen generator and the balloon inflation nozzle, and is cooled by immersion in a 32 gallon (121.1 liters) can of water. This equipment cools the hydrogen gas and causes condensation of water vapor. A manifold that permits the use of up to six chemical charges, a tee connector, an inflation nozzle, and neoprene hose that allows coupling of two generator sets are provided in the kit. All components shall be stowed in a single transit case.

3.1.1 Major components. The equipment consists of the following major components:

a. Manifold (six charge).
b. Coil condenser and tank.
c. Hose.
d. Inflation nozzle.
e. Tee connector.
f. Transit case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Material. The material shall be as specified herein and as shown on the applicable drawings. However, when a definite material is not specified, a material shall be used which will enable the equipment to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled, virgin, and reclaimed material. It is encouraged that recycled or reclaimed material, in lieu of virgin material, be used when practical provided it meets the requirements of this specification.

3.3.2 Cleaning. Material, parts and equipment shall be cleaned to remove contaminants such as corrosion products, dust, lubricating oils, mold release agents, sand, solder, flux residues, outgassing products and any other contaminant. The cleaning procedure and cleaning agents shall have no deleterious effect on the performance requirements of the materials, parts and equipments. The same cleaning procedure shall be applied to all equipments.

3.4 Design.

3.4.1 Construction. The equipment shall be constructed in accordance with SC-DL-380825, and as specified herein.

3.4.2 Seams. When tested as specified in 4.6.1, all external seams shall be gas tight and shall not leak.

3.4.3 Soldering.

3.4.3.1 Solder. The solder shall conform to QQ-S-571 or QQ-B-654 and the requirements as shown on the drawings.

3.4.3.2 Acid or acid salts. Acid or acid salts may be used in preparation for, and during soldering, including preliminary tinning and soldering of mechanical joints. Where acid or acid salts are used, they shall be completely neutralized and removed immediately after use.

3.4.3.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions and shall not be crystallized, overheated or underheated. The minimum amount of flux and solder shall be used. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles or other foreign material remaining in the equipment, flux being spread over a larger area, or damage to the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

3.4.4 Welding. The surfaces of the material to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the equipment to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.188 inch (4.8mm) unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication shall be at the option of the contractor.

3.5 Transit case. The transit case shall conform to MIL-C-4150, type, style, class, and variety as specified (see 6.2).

3.6 Shock; drop. The transit case with all components stowed and the top secured, shall withstand a free drop from a height of 48 inches (1.22cm) onto a concrete floor without damage which would impair the functioning of the case or its contents.

3.7 Preconditioning. The equipment shall be capable of meeting the requirements herein without subsequent processing after subjection to the bounce preconditioning (see 4.6.4).

3.8 Interchangeability. Like units and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.6.5).

3.9 Finish. The equipment shall be finished in accordance with MIL-F-14072, and as specified on the drawings.

3.10 Marking. Marking shall conform to MIL-M-13231 and as specified on the drawings, except that serial numbers are not required.

3.11 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following:

3.3.2 Cleaning.
3.4.3 Soldering.
3.4.4 Welding.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article inspection. The first article inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspections specified for group A, group B, and group C (see tables II, III, and IV respectively). The first article inspection shall be performed in the order shown in table I. Other first article inspections shall be performed in any order satisfactory to the Government. After completion of the first article inspection, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection.

TABLE I. First article (order of inspection).

Inspection	Requirement paragraph	Test method paragraph
1. Visual and mechanical	3.11	4.6.6
2. Shock; drop	3.6	4.6.3
3. Interchangeability	3.8	4.6.5
4. Leakage	3.4.2	4.6.1

4.3.2 First article units. When specified, the contractor shall furnish first article units of the equipment (see 6.3).

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements:

Item	Where required
Transit case	3.5
Finish	3.9
Marking	3.10
Workmanship	3.11

4.5 Quality conformance inspection of equipment before packaging.

4.5.1 Inspection. Quality conformance inspection shall be as specified in 4.4 and 4.5.2 through 4.5.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

4.5.2 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in table II.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual-mechanical	3.11	4.6.6	1	4
Leakage	3.4.2	4.6.1	1	1/

1/ All defects are considered major.

4.5.2.1 Order of inspection within group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group B inspection. Group B inspection shall be as specified in table III. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Interchangeability	3.8	4.6.5	6.5

4.5.3.1 Group B sampling. Sampling for group B shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be as specified in table III.

4.5.4 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall be as specified in table IV.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Welding	3.4.4	4.6.2
Shock, drop	3.6	4.6.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

4.5.4.1 Sampling for inspection. Two sample units from every 50 or fraction thereof produced shall be selected at random for all inspections in table IV. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.4.2 Order of inspection within group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

4.5.4.3 Group C failure. Actions required relative to group C failures shall be as specified (see 6.2).

4.6 Methods of inspections.

4.6.1 Seams. The coil condenser and trap with the outlet tubes capped, shall be completely submerged in water for 30 seconds, in order to trap air in the body of the unit. There shall be no evidence of leakage from the condenser assembly. An air hose shall be connected to the large tube of the manifold, and each of the six branches and the narrow tube leading from the large tube shall be sealed. The pressure shall be adjusted to 10 psi (68.9kPa) and the manifold immersed in water. Nonconformance to 3.4.2 shall constitute failure of this test.

4.6.2 Welding. Each unit shall be tested for maximum stress; spot or tack welds will only be tested to determine their suitability for the intended purpose. The welded joint shall not leak. Nonconformance to 3.4.4 shall constitute failure of this test.

4.6.3 Shock; drop. The free fall drop test shall be in accordance with FED-STD-101, method 5007, procedure A, except that the drop height shall be 48 inches (122cm). Nonconformance to 3.6 shall constitute failure of this test.

4.6.4 Preconditioning. The modification kit with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the package tester. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch (0.79mm) thick strip of material under one corner or edge of the modification kit to a distance of 3 inches (7.6cm) as the modification kit bounces. The modification kit shall be subjected to this test for one minute. After the test the modification kit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection. Nonconformance to 3.7 shall constitute failure of this test.

4.6.5 Inspection for interchangeability. The dimensions shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified limits, it shall be considered a major defect.

4.6.6 Visual and mechanical inspection. The modification kit shall be examined for defects listed in MIL-STD-252. Nonconformance to 3.11 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table V, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE V. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect or illegible.	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. A3-1-X-XX-XX-XX-X-XX-X-XX-A.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The modification kit MK-710( )/TMQ is intended for use with Hydrogen Generator Set AN/TMQ-3 and Meteorological Station AN/TMQ-4. The kit enables the operator to generate a greater volume of hydrogen gas in one generation cycle.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval (see 3.2).

c. When the welding process will not be at the option of the contractor (see 3.4.4).

d. Type, style, class, and variety of transit case required (see 3.5).

e. Action required relative to group C failure (see 4.5.4.3).

f. Level A or B preservation and packing (see section 5).

g. When first article inspection rough handling tests are not required.

h. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item to be tested shall be a first article sample. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Nomenclature. The parentheses in the nomenclature (see 1.1) will be deleted or replaced by a letter identifying the particular design, for example: MK-710W/TMQ. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

Custodian:
Army - ER

Preparing activity:
Army - ER

(Project 6660-A069)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provices this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY DEPARTMENT OF THE ARMY

US ARMY ELECTRONICS RESEARCH AND DEVELOPMENT COMMAND
COMBAT SURVEILLANCE AND TARGET ACQUISITION LABORATORY
ATTENTION: DELCS-PE
FORT MONMOUTH, NEW JERSEY 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-55289A(ER)	MODIFICATION KIT MK-710( )/TMQ (UNITS OF)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:42Z
Check the source to verify that this is the current version before use.